Remote Sensing Image–Text Retrieval With Implicit–Explicit Relation Reasoning

Lingling Yang, Tongqing Zhou, Wentao Ma, Mengze Du, Lu Liu, Feng Li, Shan Zhao, and Yuwei Wang

***Abstract*— Remote sensing image–text retrieval (RSITR) has become a research hotspot in recent years for its wide application. Existing methods in this context, based either on local or global feature matching, overlook the sensing variation-leaded visual deviation and geographically nearby image–text mismatching problems of remote sensing (RS) images. This work notes that this would limit the retrieval accuracy for RSITR. To handle this, we present IERR, an implicit–explicit relation reasoning framework that learns relations between local visual–textual tokens and enhances global image–text matching without requiring additional prior supervision. Specifically, masked image modeling (MIM) and masked language modeling (MLM) are used for symmetric mask reasoning consistency alignment. Meanwhile, masked features (i.e., implicit relation) and unmasked features (i.e., explicit relation) are fed into a multimodal interaction encoder to enhance the representations of the textual–visual features. Extensive experimental results on the RSICD and RSITMD datasets demonstrate the superiority of IERR compared with 17 baselines.**

***Index Terms*— Implicit relation reasoning, masked image modeling (MIM), masked language modeling (MLM), remote sensing image–text retrieval (RSITR).**

I. Introduction

As a fundamental and long-standing task in the vision-and-language domain, remote sensing image–text retrieval (RSITR) is devoted to bridging the semantic gap between the visual and textual modalities, and searching images for a given textual description or vice versa. Given its potential applications in areas including resource acquisition [1], disaster monitoring [2], and urban management [3], [4], RSITR is progressively garnering attention in both the academic and industrial communities [5], [6].

Existing RSITR methods are built either on coarse-grained alignment or fine-grained alignment. Coarse-grained global alignment [7], [8], as shown in Fig. 1(a), separately maps the global features of visual and textual samples into a joint embedding space to calculate the cross-modal similarity for retrieval purpose. In contrast, fine-grained local alignment [9], [10], [11], as shown in Fig. 1(b), models fine-grained semantic interactions by multilevel feature representations for estimating the similarity. Despite impressive progress, both categories only consider narrowing the modality gap between a specific image and a piece of text, while ignoring the similarities and correlations among multiple images and texts. In fact, it is common in RSITR benchmark datasets [11], [12] that images (or texts) with different labels show strong semantic affinity [13], [14]. In effect, such affinity between samples' intramodality brings two challenges for those RSITR methods that only rely on narrowing the modality gap.

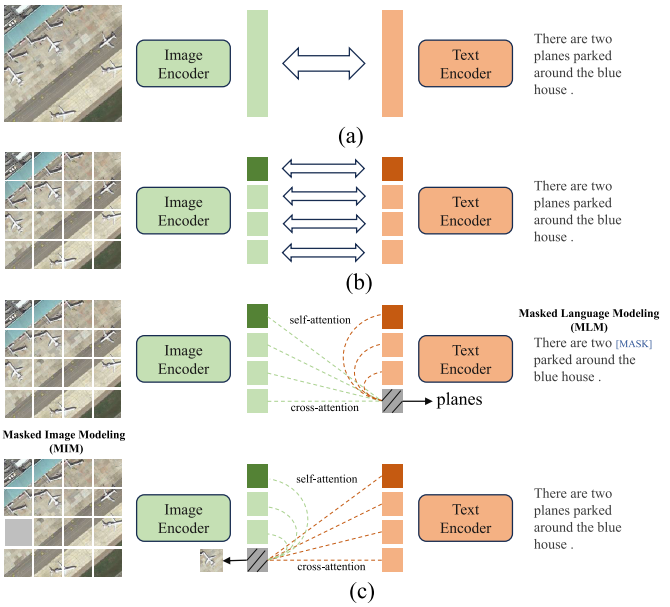


Fig. 1. Comparisons on RS image–text models in existing methods and this work. (a) Coarse-grained global alignment, (b) fine-grained local alignment, and (c) implicit–explicit relation reasoning in this work.

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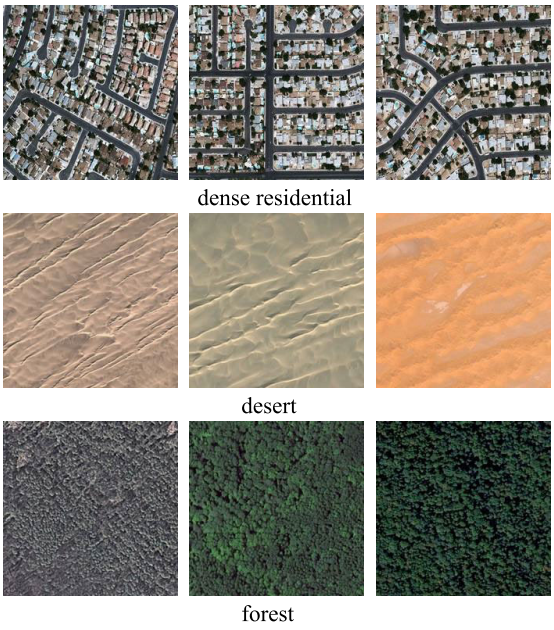


Fig. 2. Example modal diversity in image modality.

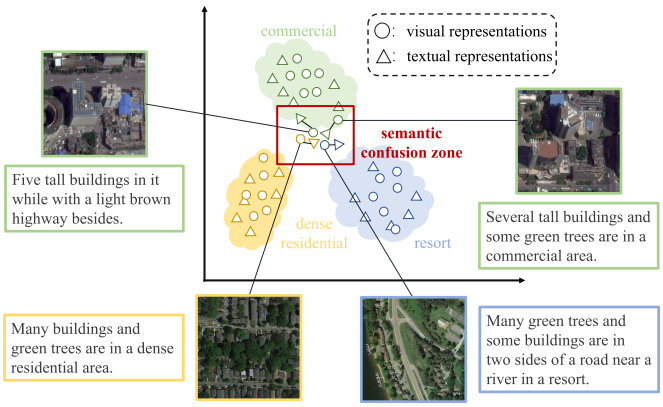


Fig. 3. Example for the distribution of image and text embeddings in the semantic space. The red box indicates the semantic confusion zone. The green, yellow, and blue circle-triangle pairs indicate image–text pairs of commercial, dense residential, and park scene, respectively.

1) *Intramodal Diversity*: Remote sensing (RS) imaging is susceptible to uncontrollable factors such as weather, visibility, and viewpoint [13], [14], [15], [16], causing significant differences in the visual representation of images under the same scene, even when they share the same label, in Fig. 2. Meanwhile, RS images show a small ratio of image foreground which makes the semantic objects in RS images insignificant [17], lack specific themes and focal points [18], which can generate various descriptions from different perspectives, encompassing different combinations and permutations of the underlying fine-grained semantic units, which makes the descriptions varied. Such fluctuations lead to semantic deviation.
2) *Intermodal Mismatching*: RS images from different locations may contain similar geographical features, with the corresponding text descriptions sharing similar semantic elements [19].

Hence, an image may be mistakenly paired with text descriptions intended for other images (i.e., semantic confusion), resulting in an ambiguous boundary in the visual–textual joint representation space, as illustrated in Fig. 3.

In light of the above analysis of the challenges, we propose a novel method of implicit–explicit relation reasoning for RSITR tasks, termed IERR, by exploring intramodal relation semantic enhancement and intermodal symmetric mask reasoning consistent alignment. Specifically, we introduce masked language modeling (MLM) [18] and masked image modeling (MIM) [20] with random token and patch masks to extract salient feature expressions from texts and images and combine them via self-attention and cross-attention mechanisms to relieve the semantic deviation (for Challenge 1). An implicit local fine-grained information enhancement mechanism is designed, which feeds masked features extracted from MIM and MLM and unmasked features from original images or texts into a multimodal interaction encoder, allowing for better discrimination between samples within the same label (for Challenge 2). In this way, IERR establishes consistent alignment between RS images and texts, rendering refined visual–textual joint representation.

The main contributions of this work are as follows.

1) We propose a novel implicit–explicit relation reasoning model, which is, to the best of our knowledge, the first attempt to dig implicit and explicit relation in textual and visual features at RS image–text cross-modal retrieval tasks.
2) We design two modules, relation semantic enhancement and symmetric mask reasoning consistent alignment, to ensure a refined and discriminative visual–textual joint representation space, bridging the semantic gap between images and texts in RS.
3) On two public benchmarks of RSITR, RSICD, and RSITMD, we evaluate the IERR via the comparison with 17 state-of-the-art (SoTA) baselines and a series of ablation studies. Both quantitative and qualitative experimental results demonstrate that our proposal can yield promising performance.

In the rest of this article, we review previous works related to RSITR in Section II. Then, the proposed method is introduced in Section III. The experimental results and discussion are given in Sections IV and V. Finally, we summarize in Section VI.

II. Related Work

In this section, we will briefly review the most relevant study to our work, including the different fields of RS images, RSITR, and the vision-language pretraining (VLP) model.

A. Different Fields of RS Images

RS images contain high semantic information from various scenes, making them a prominent research area in RS. Several subfields within RS imagery have garnered significant attention, including RS scene classification (RSSC) [13], [14], [21], RS image object recognition (RSIOR) [22], [23], [24], [25],

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and RS image semantic segmentation (RSIMSS) [26], [27], [28], [29]. In addition, some AI models [4], [30], [31] and other related tasks have emerged as important areas of study.

In the field of RSIMSS, Hong et al. [26] introduced a novel set of multimodal RS benchmark datasets specifically designed for cross-city semantic segmentation. This dataset includes scenes from Berlin–Augsburg and Beijing–Wuhan, facilitating research on cross-city issues. They also developed an innovative high-resolution domain adaptation network tailored for these cross-city scenarios.

Moreover, in the domain of multimodal RS, cross-modal image–image retrieval also has received widespread attention. This retrieval process involves identifying target RS images using a query image from different origions [32], different sensors [33], or even a sketch image [34]. For instance, Li et al. [32] employed deep hashing convolutional neural networks (CNNs) to enhance image retrieval. Chaudhuri et al. [33] proposed a method for cross-modal retrieval between panchromatic (PAN) and multispectral imagery. Furthermore, a multiscale model incorporating CNNs and fully connected layers was designed for retrieving RS images using a coarse sketch [34].

B. RS Image–Text Retrieval

Recent advancements in the task of RSITR have resulted in the development of two primary approaches: two-stage approaches based on caption matching and single-stage embedded-based methods [35].

The caption-based method is a two-stage approach that first converts RS images into textual descriptions and then calculates the similarity between the generated texts and the query texts. Shi and Zou [36] proposed an automatic subtitle generation framework for RS images to produce robust RS captions. Similarly, Lu et al. [12] provided a publicly available RS image–text dataset and proposed a method for automatic RS image subtitle generation and image–text cross-modal retrieval using subtitles. However, these two-stage methods suffer from performance issues due to information loss at each stage.

With the progression of deep learning, single-stage embedded-based methods have gained prominence. These methods embed RS images and query texts into the same semantic space and then calculate their similarity. Abdullah et al. [37] introduced a deep bidirectional triplet network (DBTN) to enable learning more robust representations. To address multiscale scarcity and target redundancy in RSITR, Yuan et al. [11] proposed an asymmetric multimodal feature matching network (AMFMN) and created a fine-grained RS image–text dataset. Cheng et al. [38] developed a semantic alignment module to obtain more discriminative feature representation by exploring the potential correspondence between RS images and text. Lv et al. [39] suggested a fusion-based correlation learning model for RS image–text retrieval and solved the heterogeneity gap problem through knowledge distillation. From another perspective, Yuan et al. [35] introduced a lightweight text–image retrieval model to achieve faster RS cross-modal retrieval, employing knowledge distillation and contrast learning to enhance the retrieval performance. Zhang et al. [40] proposed a novel alignment based on curriculum learning and designed a key-entity attention (KEA) mechanism to address information imbalance across different modalities. To mitigate semantic confusion in RS, Pan et al. [17] designed a scene-aware aggregation network (SWAN) to improve the scene perception capability. Yuan et al. [10] used the CLIP pretraining model, achieving promising results in RSITR. Zhang et al. [18] employed MLM to capture finer-grained image–text correlations. In addition, Zhang et al. [20] proposed a novel framework based on masked-guided relation modeling to fully explore intramodal and intermodal relations, which inspires us a lot.

C. Vision-Language Pretraining

VLP aims to understand the semantic relationships between visual and language modalities by leveraging extensive image–text pair datasets. Inspired by the successes of Transformer-based models such as BERT [41], and vision transformer (ViT) [42], VLP has become crucial for learning multimodal representations, showing strong performance in tasks like image captioning [43], image–text retrieval [44], and visual question answering [45]. VLP models can be typically classified into two categories based on their model structure: single-stream and dual-stream models.

Single-stream models (e.g., OSCAR [46], XGPT [47], and SimVLM [48]) combine textual and visual features into one transformer encoder. This method is parameter-efficient as it uses a unified set of parameters for both types of data. However, these models can be slow during inference since they must calculate similarity scores for numerous image–text pairs.

Dual-stream models use two separate encoders to extract textual and visual features, operating independently without shared parameters. Examples include ViLBER [49] and ALBEF [50]. These models excel in image–text retrieval but are less effective at handling complex interactions between vision and language. Contrastive language-image pretraining (CLIP) [51] employs a contrastive loss function to align similar image–text pairs closely in the representation space, effectively establishing strong connections between images and text. The training process of CLIP involves a large-scale dataset comprising images and text, including noisy data from the Internet [52], [53], [54]. Through this approach, it learns universal representations of image–text features, enabling impressive performance across various downstream tasks, i.e., text–video retrieval [55], [56], [57], text-to-image person retrieval [57], [58], [59], [60], action localization [61], RSITR [10], [52]. CLIP has set new standards in image–text retrieval, significantly contributing to the advancement of VLP.

III. Method

In this section, our proposed method IERR framework is introduced. The structure of IERR is shown in Fig. 4. The visual and textual representations are described in detail in Section III-A. Implicit relation reasoning and the loss function are introduced in Sections III-B and III-C.

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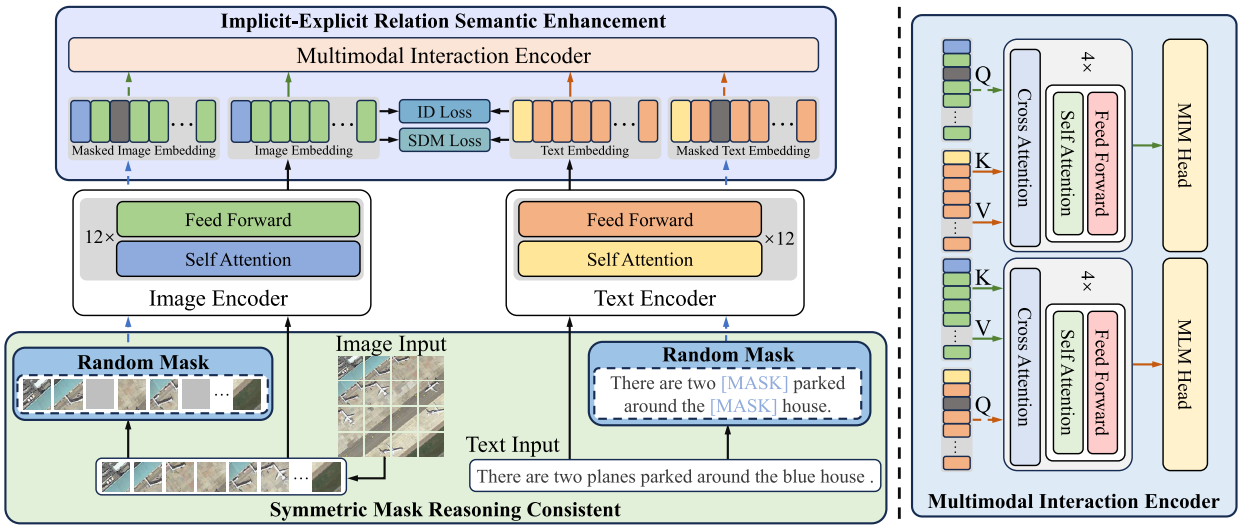


Fig. 4. Structure of IERR framework. It consists of implicit–explicit relation semantic enhancement and symmetric mask reasoning consistency.

A. Visual–Textual Feature Representation

Previous works in cross-modal text–image retrieval are more likely using text and image encoders which are pretrained separately on unimodal larger-scale datasets. Inspired by [45], the full CLIP image and text encoder are adopted to get the visual and textual representations.

The image encoder is a CLIP pretrained ViT model to obtain the image embedding. The input image $I \in R^{H\times W\times C}$ is split into a sequence of $N_v = H \times W/P^2$ pixel nonoverlapping patches, where *(H, W)* is the resolution of the original image and P is the patch size. Then, the patch sequence is mapped into 1-D tokens $\{h_i^V\}|_{i=1}^{N_v}$ by a trainable linear projection, where h_i^V denotes the feature of the ith patch, and N_v is the number of patches. After the linear projection, the positional embedding and the learnable classification labels are embedded into the special [CLS] token. The visual encoding output is $h^V = \{h_{\text{cls}}^V, h_1^V, \ldots, h_{N_v}^V\}$, where h_{cls}^V is the classfication label feature.

The text encoder is the CLIP text encoder to extract the text representation. Following CLIP, the lower cased byte pair encoding (BPE) with a 49 152 vocab size is used to tokenize the input text description T. The text description is backend with [SOS] and [EOS] tokens to indicate the start and the end of the sequence. After feeding into the transformer, the textual output is $h^T = \{h_{\text{sos}}^T, h_1^T, \ldots, h_{N_t}^T, h_{\text{eos}}^T\}$, where N_t is its token length.

B. Design of IERR

1) Symmetric Mask Reasoning Consistent: MLM has become widely known in the field of natural language processing (NLP), like BERT [62]. MIM is proposed in computer vision (CV), like BEiT [63]. Similar to the analysis of Fu et al. [64], there is an advantage to the usage of MIM and MLM which is eliminating unnecessary and redundant information about each modality. Both MIM and MLM are used in our method, and these two models consist of the symmetric mask reasoning consistent.

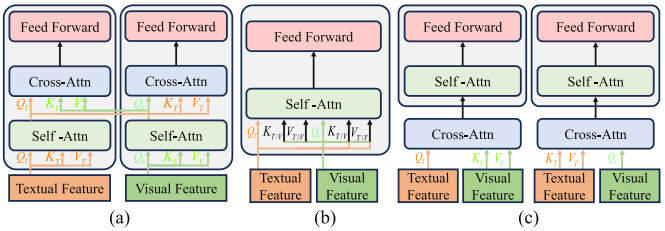


Fig. 5. Illustration of three popular interaction modules. (a) Co-attention, textual, and visual features are fed into separate transformer blocks with self-attn and cross-attn independently. (b) Merged attention, textual, and visual features are concatenated together and then fed into a single transformer block. (c) Multimodal attention, textual, and visual features are first fused by a cross-attn layer and then fed into a single transformer block.

Given an input image I, we randomly mask out the image patches with a probability of 20% and replace them with special noise. For an input text description T, we randomly mask out the text tokens with a probability of 15% and replace them with the special token [MASK]. Then, the masked image $\hat{I}$ and masked text $\hat{T}$ are fed into the image and text encoders, respectively. h^{V_m} and h^{T_m} are used as masked visual and masked textual features. And from the last hidden states, $\{h_i^{V_m}\}_{i=1}^{N_v}$ and $\{h_j^{T_m}\}_{j=1}^{N_t}$ are used as masked visual and masked textual features.

2) Implicit–Explicit Relation Semantic Enhancement: Following the work of Jiang and Ye [58], we use the multimodal interaction encoder to fuse the image and text embedding, as shown in Fig. 5. There are four-layer transformer blocks and a multihead cross in the multimodal interaction encoder which is an important component of implicit relation semantic enhancement.

For the multimodal interaction encoder, the masked feature ($\{h_i^{V_m}\}_{i=1}^{N_v}$ / $\{h_j^{T_m}\}_{j=1}^{N_t}$) is served as query ($Q$), and the unmasked feature ($\{h_i^{V}\}_{i=1}^{N_v}$ / $\{h_j^{T}\}_{j=1}^{N_t}$) is fed as key (K) and value (V). The full interaction between text and masked image representations, and image and masked text can be achieved by

$$\{h_j^{T_{M_t}}\}_{j=1}^{N_t} = \text{Transformer}(\text{MCA}(\text{LN}(Q, K, V))) \quad (1)$$

$$\{h_i^{T_{M_v}}\}_{i=1}^{N_v} = \text{Transformer}(\text{MCA}(\text{LN}(Q, K, V))) \quad (2)$$

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where $\{h_j^{T_{M_t}}\}_{j=1}^{N_t}$ is the fused masked image ($\{h_i^{V_m}\}_{i=1}^{N_v}$) and text ($\{h_j^{T}\}_{j=1}^{N_t}$) contextualized representations, $\{h_i^{T_{M_v}}\}_{i=1}^{N_v}$ is the fused image ($\{h_i^{V}\}_{i=1}^{N_v}$) and masked text ($\{h_j^{T_m}\}_{j=1}^{N_t}$) contextualized representations. LN$(\cdot)$ denotes layer normalization, the MCA$(\cdot)$ is the multihead cross attention and can be realized by

$$\mathrm{MCA}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{3}$$

where d is the embedding dimension of masked tokens.

For each masked position $\{h_i^{T_{M_v}} : i \in |\mathcal{M}_t|\}_{i=1}^{N_v}$, multilayer perception(MLP) classifier is used to predict where $\{m_{i,j}^{T_m}\}_{j=1}^{|\mathcal{V}|} = \mathrm{MLP}(h_i^{T_{M_v}})$, $\{m_{i,j}^{T_m}\}_{j=1}^{|\mathcal{V}|}$ is the probability of ith visual token in $\mathcal{V}$ of jth masked textual token. The MLM objective can be formulated as

$$\mathcal{L}_{\mathrm{mlm}} = -\frac{1}{|\mathcal{M}_t||\mathcal{V}|}\sum_{i\in\mathcal{M}_t}\sum_{j\in|\mathcal{V}|} y_j^i \log\frac{\exp(m_{i,j}^{T_m})}{\sum_{k=1}^{|\mathcal{V}|}\exp(m_{i,k}^{T_m})} \tag{4}$$

where $|\mathcal{M}_t|$ denotes the set of masked textual tokens and $\mathcal{V}$ is the text vocabulary. y_j^i is 1 if the true label of ith masked tokens is jth vocabulary in $\mathcal{V}$, and 0 otherwise.

For each masked position $\{h_j^{T_{M_t}} : j \in |\mathcal{M}_v|\}_{j=1}^{N_t}$, multilayer perception (MLP) classifier is used to predict the probability of the corresponding original tokens $\{m_{i,j}^{V_m}\}_{j=1}^{|\mathcal{C}|} = MLP(h_j^{T_{M_t}})$. The MIM objective can be formulated as

$$\mathcal{L}_{\mathrm{mim}} = -\frac{1}{|\mathcal{M}_v||\mathcal{C}|}\sum_{i\in\mathcal{M}_v}\sum_{j\in|\mathcal{C}|} y_j^i \log\frac{\exp(m_{i,j}^{V_m})}{\sum_{k=1}^{|\mathcal{C}|}\exp(m_{i,k}^{V_m})} \tag{5}$$

where $|\mathcal{M}_v|$ denotes the set of masked image tokens and $\mathcal{C}$ is the label set for tokens. y_j^i is 1 if the true label of ith masked image tokens is jth label in $\mathcal{C}$, and 0 otherwise. $\{m_{i,j}^{V_m}\}_{j=1}^{|\mathcal{C}|}$ is the probability of jth word in $\mathcal{C}$ of ith masked image token.

The final $\mathcal{L}_{\mathrm{irse}}$ is

$$\mathcal{L}_{\mathrm{irse}} = \mathcal{L}_{\mathrm{mim}} + \mathcal{L}_{\mathrm{mlm}} \tag{6}$$

C. Loss Function

Following previous work, the cross-modal matching loss incorporates the cosine similarity distributions of the image–text pairs embeddings into KL divergence to associate the representations across different modalities.

Given a mini-batch of N image–text pairs, for each image global representation $h_{\mathrm{cls},i}^V$ and each sentence global representation $h_{\mathrm{sos},j}^T$, we construct a set of image–text representation pairs as $\{(h_{\mathrm{cls},i}^V, h_{\mathrm{sos},j}^T), y_{i,j}\}_{j=1}^N$, where $y_{i,j}$ is a true matching label, $y_{i,j} = 1$ means that $(y_{i,j})$ is a matched pair from the same identity, while $y_{i,j} = 0$ indicates the unmatched pair. Then, the probability of matching pairs can be simply calculated with the following softmax function:

$$p_{i,j} = \frac{\exp(\mathrm{sim}(h_{\mathrm{cls},i}^V, h_{\mathrm{sos},j}^T)/\tau)}{\sum_{k=1}^N \exp(\mathrm{sim}(h_{cls,i}^V, h_{\mathrm{sos},k}^T)/\tau)} \tag{7}$$

where N is mini-batch size, τ is a temperature hyperparameter which controls the probability distribution peak

$$\mathcal{L}_{i2t} = \mathrm{KL}(p_i \parallel q_i) = \frac{1}{N}\sum_{i=1}^N\sum_{j=1}^N p_{i,j}\log\left(\frac{p_{i,j}}{q_{i,j}+\varepsilon}\right) \tag{8}$$

$$\mathcal{L}_{\mathrm{sdm}} = \mathcal{L}_{i2t} + \mathcal{L}_{t2i} \tag{9}$$

$$\mathcal{L}_{id} = -(y_{id}\log(\mathrm{softmax}(W_{id}v_{\mathrm{cls}})) + y_{id}\log(\mathrm{softmax}(W_{id}t_{\mathrm{cls}}))) \tag{10}$$

where W_{id} is a shared transformation matrix to classify the different figures and y_{id} is the ground true identity.

The final overall optimization objective is defined as

$$\mathcal{L} = \lambda_1\mathcal{L}_{\mathrm{sdm}} + \lambda_2\mathcal{L}_{id} + \lambda_3\mathcal{L}_{\mathrm{irse}} \tag{11}$$

where $\lambda_1 + \lambda_2 + \lambda_3 = 1$.

IV. Experimental Settings and Baselines

In this section, we introduce the benchmark datasets, evaluation metrics, implementation details, and baselines.

A. Datasets

In the experiments, two benchmark datasets, RSICD [12] and RSITMD [11], are used to evaluate the performance of the proposed method. Each dataset consists of images accompanied by five descriptive sentences. The RSICD dataset comprises 10 921 images with a resolution of 224 × 224 pixels. There are 4743 images with a size of 256 × 256 pixels in the RSITMD dataset. The descriptions in the RSITMD dataset exhibit a finer level of granularity compared to RSICD. To ensure consistent evaluation, the datasets are split into three subsets following the setting proposed in [65]. Specifically, the datasets are divided into 80% for training, 10% for validation, and 10% for testing. This partitioning scheme allows for fair and refined comparisons of the performance of the method with existing approaches on both the RSICD and RSITMD datasets.

B. Metrics

In accordance with previous RSITR studies, the effectiveness of our method is evaluated using the metrics of $R@K$ ($K = 1, 5, 10$) and mR. The $R@K$ metric assesses the percentage of ground-truth matched pairs among the top K ranked results. The mR metric, on the other hand, represents the average of the $R@K$ values, providing an overall evaluation of the retrieval performance.

C. Implementation Details

All experiments are carried out on a single RTX4080 Ti 16 GB GPU. During training, image data augmentation techniques including random horizontal flipping, random crop with padding, and random erasing are employed. The input images are resized to a dimension of 256 × 256 pixels. The maximum length of the textual token sequence is set to 77. The model optimizer used for training is Adam, with a total of 60 epochs. The learning rate is initialized to 1×10^{-5}, and a gradual increase from 1×10^{-6} to 1×10^{-5} is applied over the first 5 warm-up epochs. For randomly initialized modules, the initial learning rate is set to 5×10^{-5}. The temperature parameter is fixed at 0.02.

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TABLE I
COMPARISONS OF CROSS-MODAL RETRIEVAL RESULTS OF RSICD AND RSITMD DATASETS

Method	RSICD							RSITMD						
	I2T			T2I			*mR*	I2T			T2I			*mR*
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++ [7]	3.38	9.51	17.46	2.82	11.32	18.10	10.43	10.38	27.65	39.6	7.79	24.87	38.67	24.83
SCAN t2i [66]	4.39	10.9	17.64	3.91	16.20	26.49	13.25	10.18	28.53	38.49	10.10	28.98	43.53	26.64
SCAN i2t [66]	5.85	12.89	19.84	3.71	16.40	26.73	14.23	11.06	25.88	39.38	9.82	29.38	42.12	26.82
CAMP-triplet [67]	5.12	12.89	21.12	4.15	15.23	27.81	14.39	11.73	26.99	38.05	8.27	27.79	44.34	26.20
CAMP-bce [67]	4.20	10.24	15.45	2.72	12.76	22.89	11.38	9.07	23.01	33.19	5.22	23.32	38.36	22.03
MTFN [68]	5.02	12.52	19.74	4.90	17.17	29.49	14.81	10.4	27.65	36.28	9.96	31.37	45.84	26.92
LW-MCR(b) [35]	4.57	13.71	20.11	4.02	16.47	28.23	14.52	9.07	22.79	38.05	6.11	27.74	49.56	25.55
LW-MCR(d) [35]	3.29	12.52	19.93	4.66	17.51	30.02	14.66	10.18	28.98	39.82	7.79	30.18	49.78	27.79
AMFMN-soft [11]	5.05	14.53	21.57	5.05	19.74	31.04	10.78	11.06	25.88	39.82	9.82	33.94	51.9	28.74
AMFMN-fusion [11]	5.39	15.08	23.40	4.90	18.28	31.44	16.42	11.06	29.20	38.72	9.96	34.03	52.96	29.32
AMFMN-sim [11]	5.21	14.72	21.57	4.08	17.00	30.60	15.53	10.63	24.78	41.48	11.51	34.69	54.87	29.66
GaLR [65]	6.50	18.91	29.70	5.11	19.57	31.92	16.62	13.05	30.09	42.70	10.47	36.34	53.35	31.00
GaLR-MR [65]	6.59	19.85	31.04	4.69	19.48	32.13	18.96	14.82	31.64	42.48	11.15	36.68	51.68	31.41
CLIP [51]	7.96	21.41	32.48	5.42	20.02	33.19	20.08	14.89	33.55	45.64	9.81	32.68	47.43	30.67
HVSA [40]	7.47	20.62	32.11	5.51	21.13	34.13	20.16	13.20	32.08	45.58	11.43	39.2	57.45	33.16
DOVE [19]	8.66	22.35	34.95	6.04	23.95	40.35	22.72	16.81	36.8	49.93	12.20	49.93	66.50	37.73
MSITA [69]	8.67	22.71	33.91	6.13	21.98	35.39	21.47	15.22	34.20	47.65	12.15	39.92	57.72	34.48
IERR	**17.29**	**35.41**	**48.58**	**13.07**	**35.32**	**51.31**	**33.50**	**27.66**	**48.89**	**61.06**	**24.43**	**56.59**	**72.66**	**48.55**

D. Baselines

The proposed method IERR is compared with several SoTA approaches in the field of RSITR on RSICD and RSITMD datasets. The results are shown in Table I.

1) *VSE++ [7]:* VSE++ focuses more on difficult on differenct sample pairs. The alignment of cross-modal features is measured by the MH loss function.
2) *SCAN [66]:* SCAN presents stacked cross attention to discover the full latent alignments using both image regions and words in a sentence as context and infer image–text similarity.
3) *CAMP [67]:* CAMP proposes cross-modal adaptive messaging, which adaptively controls the flow of information for cross-modal messaging.
4) *MTFN [68]:* MTFN utilizes rank-based tensor visual–textual fusion features to learn the image–text similarity.
5) *LW-MCR [35]:* LW-MCR proposes a hidden supervised optimization method based on knowledge distillation to improve the accuracy.
6) *AMFMN [11]:* AMFMN uses multiscale image features to guide text features' representation, and the query incorporates keywords in addition to text.
7) *GaLR [65]:* GalR proposes a multiscale information dynamic fusion module to integrate global and local visual features, which are extracted by graph neural networks and CNNs.
8) *CLIP [51]:* CLIP is a cross-modal pretrained model designed for contrast-based image–text learning.
9) *HVSA [40]:* HVSA uses an adaptive alignment strategy based on curriculum learning which aligns the RS image–text paris from easy to hard.
10) *DOVE [19]:* DOVE uses regional visual features to adjust the distance between visual and textual features.
11) *MSITA [69]:* MSITA designs an IGTA module to provide image information to guide text alignment, which enables the effective capture of fine-grained correspondences between RS image regions and textual descriptions.

V. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we conduct extensive experiments to answer the following research questions (RQs).

RQ1: Is the overall performance of IERR superior to the baselines?

RQ2: How does the design of each component in our model affect the retrieval performance?

RQ3: What is the result of qualitative visualization?

RQ4: What are the hyperparameters of the loss function?

A. Comparision With SoTA (RQ1)

Our approach surpasses existing methods across all metrics in this experiment. For image-to-text retrieval, IERR achieves R@1, R@5, and R@10 values of 15.37%, 34.68%, and 47.21%, respectively, on the RSICD dataset. For text-to-image retrieval, the corresponding metrics are 11.66%, 31.82%, and 46.02%. Compared to the CLIP model, our method shows enhancements of 7.41%, 13.27%, and 14.73% for R@1, R@5, and R@10 in image-to-text retrieval, and 6.24%, 11.80%, and 12.83% in text-to-image retrieval. Against previous work, there is a 9.65% improvement in *mR*, reaching 31.12% compared to MSITA. The strong performance on the RSICD dataset underscores the effectiveness of our alignment strategy.

On the RSITMD dataset, our method also achieves the leading results. Compared to the previous excellent top-performing work GaLR, our IERR approach also exhibits considerable performance improvements. For text-to-image retrieval, there are increases of 11.52%, 16.68%, and 19.52% in R@1, R@5, and R@10, respectively. For image-to-text retrieval, the improvements are 11.35%, 15.00%, and 18.73%. These outcomes highlight the superior fine-grained interaction between text and images and the generalization ability of our model.

In summary, our IERR approach consistently achieves the best performance across all metrics on both benchmark datasets, demonstrating robustness and broad applicability.

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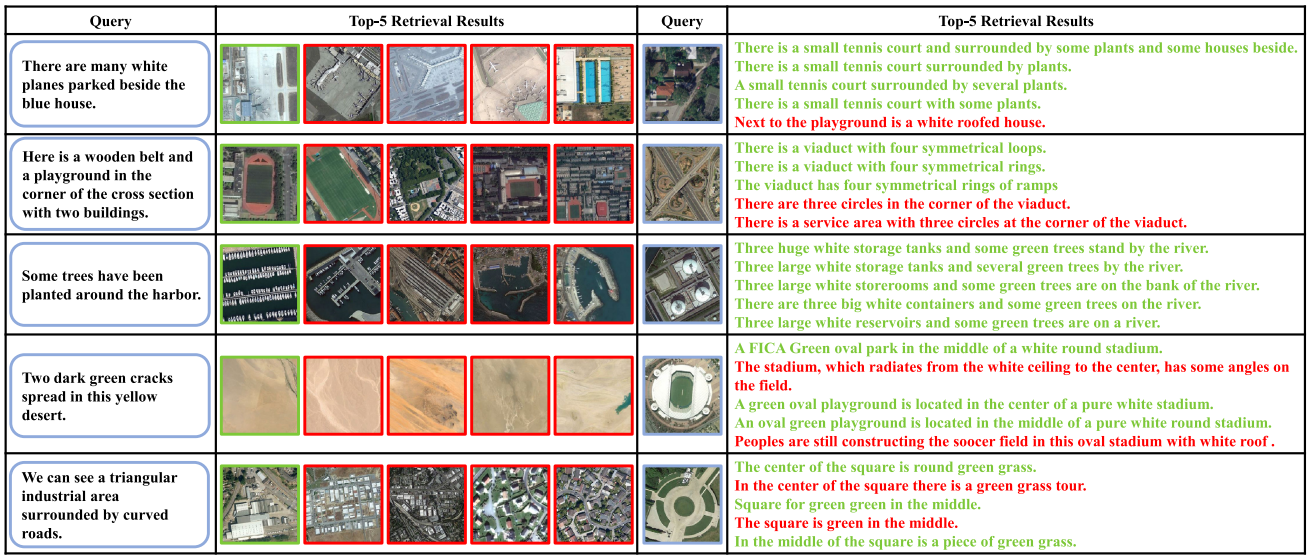


Fig. 6. Qualitative top-5 text-to-image and image-to-text retrieval results of our IERR method on the RSITMD dataset. Images with green bounding boxes are the ground truth and the ground-truth texts are marked with green. The blue boxes of texts and images are queries.

TABLE II
ABLATION STUDY IN EACH COMPONENT OF IERR ON RSICD AND RSITMD

No.	Method	RSICD							RSITMD						
		I2T			T2I			*mR*	I2T			T2I			*mR*
		R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
0	Baseline	15.10	34.58	48.03	12.55	32.77	49.68	32.12	24.78	44.25	57.74	20.75	53.14	71.68	45.39
1	SDM	14.00	32.57	44.74	11.33	31.66	46.99	30.21	24.12	42.26	54.43	19.69	49.56	69.07	43.19
2	ID	7.50	22.60	33.94	5.25	19.96	32.81	20.35	12.39	30.75	46.90	9.34	39.2	56.81	32.57
3	IRR	3.11	9.70	15.28	4.06	13.80	23.00	11.49	4.78	15.49	23.89	7.39	22.66	36.81	18.52
4	SDM+ID	16.01	33.85	48.40	13.25	**35.59**	50.41	32.92	27.21	47.79	**61.50**	24.03	53.89	68.94	47.23
5	SDM+IRR	16.83	34.22	32.03	11.95	33.25	49.08	32.99	24.78	47.79	61.28	**24.87**	52.97	67.97	46.61
6	IERR	**17.29**	**35.41**	**48.58**	**13.07**	35.32	**51.31**	**33.50**	**27.66**	**48.89**	61.06	24.43	**56.59**	**72.66**	**48.55**

B. Ablation Studies (RQ2)

In this section, several experiments are used to demonstrate the effectiveness of each different component in our IERR. The experimental results are summarized in Table II. The baseline model consists of the CLIP-ViT-B/16 model fine-tuned with the original CLIP loss (InfoNCE loss).

To demonstrate the effectiveness of each component of the loss, we evaluate each one individually on the two datasets. The SDM loss plays a crucial role in both retrieval tasks, with results closely aligning with those of the original loss. This indicates that the SDM loss effectively aligns feature representations between different modalities.

The ID and IRSE losses are used to fine-tune the network, as shown in No. 4 and No. 5 in Table II. In the RSICD dataset, combining SDM with ID improves R@1, R@5, and R@10 by 2.01%, 1.28%, and 3.66%, respectively, for the image-to-text retrieval task. For the text-to-image retrieval, the improvements are 1.92%, 3.93%, and 3.42%, respectively. The *mR* also increases by 2.71%, achieving the best R@5 and R@10 in the text-to-image retrieval task at 35.59% and 50.41%, respectively. In the RSITMD dataset, this combination achieves the highest R@5 for text-to-image retrieval at 53.89% and the best R@1 for image-to-text retrieval at 27.21%. These results demonstrate the beneficial impact of ID and IRSE on cross-modal matching.

The final loss, combining SDM, ID, and IRSE, achieves the best performance on both datasets. In the image-to-text retrieval task, it achieves R@1, R@5, and R@10 values of 17.29%, 35.41%, and 48.58%, respectively. For text-to-image retrieval, and 13.07%, 35.32% and 51.31%. The *mR* is highest at 33.50% for the RSICD dataset. For the RSITMD dataset, the *mR* reaches 48.55%, with R@1 for image-to-text at 27.66%, and R@5 and R@10 for the text-to-image at 56.59% and 72.66%. These results demonstrate the effectiveness of the combined SDM, ID, and IRSE losses in cross-modal retrieval tasks.

To demonstrate the effectiveness of the proposed implicit relation semantic enhancement, we compare it with two other popular multimodal interaction modules, co-attention [70] and merged attention [71] in Table III. Although co-attention performs the best, it also has the highest parameter count. Compared to the merged attention, the parameters of multi-interaction are more than 1.05M, and the *mR* is 0.24 higher.

C. Qualitative Results (RQ3)

We implement three qualitative experimental results, retrieval visualization, feature representation visualization, and heatmap visualization, to visually illustrate the effectiveness of our model.

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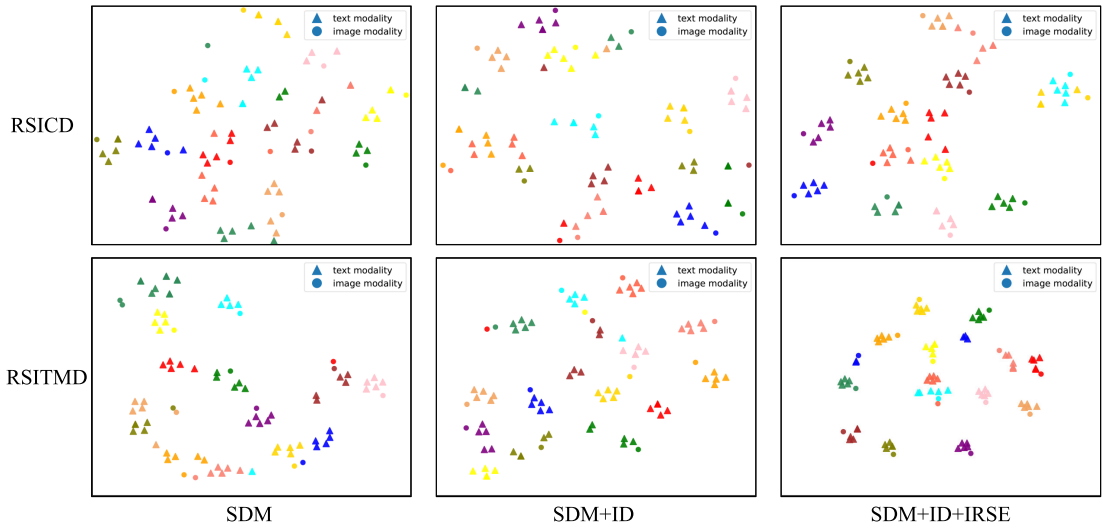


Fig. 7. Detailed change in t-SNE visualizations of RS images and texts. The circles represent image modality, and the triangles represent text modality.

TABLE III
COMPARISONS BETWEEN DIFFERENT MULTIMODAL INTERACTION MODULE OF IERR ON RSITMD

Method	Param (M)	I2T			T2I			*mR*
		R@1	R@5	R@10	R@1	R@5	R@10	
Co-attention	39.27	26.55	50.22	**63.27**	**24.51**	56.02	71.89	**48.74**
Merged attention	12.61	**28.32**	**50.51**	61.95	24.34	54.51	70.09	48.31
Multimodal attention	13.66	27.66	48.89	61.06	24.43	**56.59**	**72.66**	48.55

For retrieval visualization, Fig. 6 presents the qualitative results of image retrieval when given a sentence query and text retrieval when given an RS image query across the two datasets. Each query is accompanied by the top-5 retrieval results. In the image and text retrieval task, ground truth is highlighted in green. Fig. 6 showcases ten different scenes, namely, airport, playground, port, desert, industrial, playground, viaduct, storage tank, stadium, and square. Our model accurately distinguishes similar samples and retrieves the correct results. The proposed method demonstrates the ability to comprehend both abstract phrases and complex sentences, showcasing robustness in handling both simple and complex images.

For feature representation visualization, the t-distributed stochastic neighbor embedding (t-SNE) [72] is employed to provide a more intuitive representation of the distribution of multimodal features. t-SNE is a similarity data visualization method that preserves both local structures of the data and reveals important global structures. To ensure a fair comparison, we select 15 images and their corresponding 75 captions from the test set of the RSICD and RSITMD datasets, maintaining consistency across all methods. Fig. 7 illustrates the t-SNE visualization, showing the fusion of image embeddings and text embeddings. In the t-SNE, images and their corresponding captions with the same color represent the same image-caption pair. When only the SDM loss is used, the distributions of the text modality and image modality are more scattered. As the number of losses increases, the captions corresponding to the same image become closer to that image which indicates the effectiveness of the proposed method. This fusion is particularly evident in the RSITMD dataset.

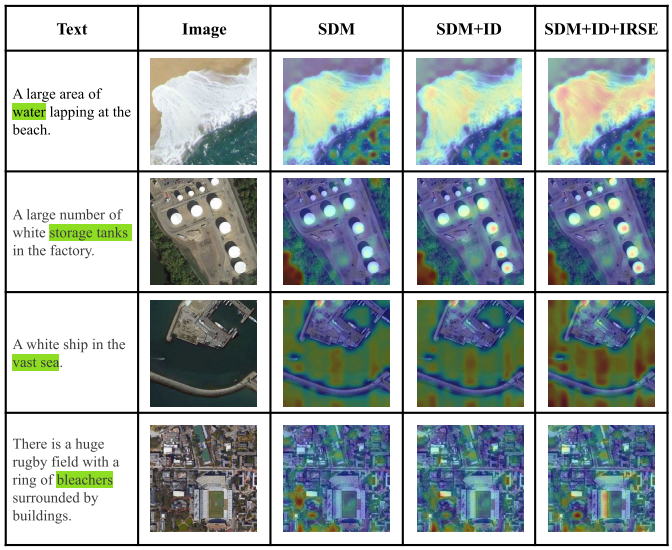


Fig. 8. Visualization of heatmap for different combinations of losses in IERR.

To better visualize the effects of the three different losses, heatmap visualization of the original images is also presented, as shown in Fig. 8. As the number of losses increases, the red regions in the heatmaps correspond more closely to the text. For the first image, the heatmap generated using all losses focuses more on "water"; for the second image, it emphasizes "storage tanks"; for the third image, it highlights "vast sea"; and for the fourth image, it emphasizes "bleachers." This additional visual representation helps illustrate the alignment between the image content and the corresponding textual information, further supporting the effectiveness of our method.

Overall, these experimental results provide evidence that the proposed IERR method has the capability to emphasize salient regions within RS images.

D. Sensitivity Analysis of Loss Function (RQ4)

We conduct several experiments to explore the relationship between the weight hyperparameters for the three loss functions. It is found that when λ_1 is held constant, a larger value

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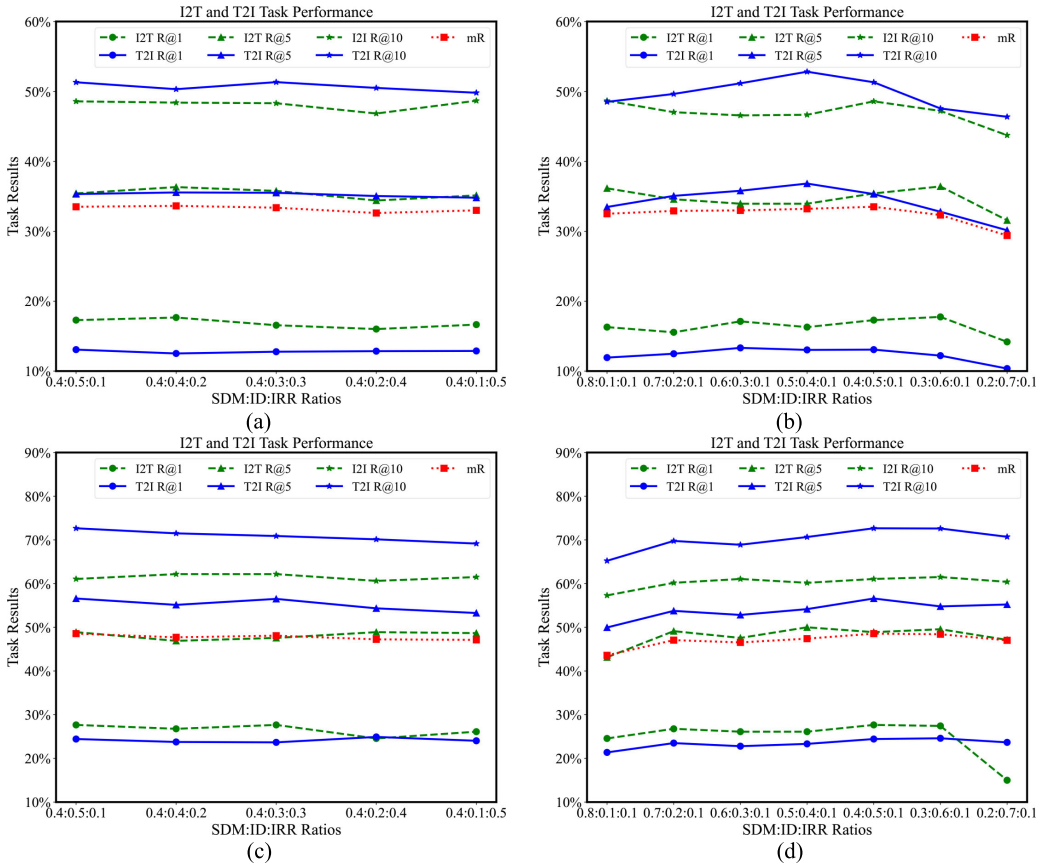


Fig. 9. Comparison of hyperparameters for loss function weights. (a) and (b) Conducted on the RSICD dataset. (c) and (d) Conducted on the RSITMD dataset. The hyperparameter λ_1 is 0.4 in (a) and (c), while λ_2 and λ_3 are variable.

of λ_2 improves retrieval performance. As shown in Fig. 9(a) and (c), the λ_1 is set to 0.4 while the weight of ID loss λ_2 varies among { 0.5, 0.4, 0.3, 0.2, 0.1} and IRSE loss λ_3 varies among { 0.1, 0.2, 0.3, 0.4, 0.5} . Fig. 9(b) and (d) shows that the best performance is achieved when λ_1, λ_2, and λ_3 are set to 0.4, 0.5, and 0.1, respectively.

VI. Conclusion

In this article, we present an IERR network that uses MIM and MLM to learn more local visual and textual local features. Extensive experiments are conducted on two classical datasets, RSICD and RSITMD, to evaluate the effectiveness of the proposed model. The results demonstrate that our approach surpasses several previous advanced methods. We also perform visualization experiments to verify our model's interpretability, demonstrating its efficiency and generalization through the t-SNE algorithm and attention heatmaps' visualization.

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